RAK19009 WisBlock Mini Base Board with Power Slot Datasheet

Overview

Description

RAK19009 is a **WisBlock Base** board that connects **WisBlock Core**, **WisBlock Power Module**, and **WisBlock Modules.** It has one slot reserved for the WisBlock Core module, one slot for the WisBlock Power module, and two Slot C-D for WisBlock Modules. The WisBlock Core and the Wisblock Power module are attached on the top side and the WisBlock Modules are attached to the bottom side of the RAK19009 WisBlock Base board. Slot C holds modules up to 23 mm in size, while Slot D supports 10 mm WisBlock Modules. Also, there are two **2.54 mm pitch headers** for extension interface with **BOOT**, **I2C**, and **UART** pins.

WisBlock Modules are connected to the RAK19009 WisBlock Base board via **high-speed board to board connectors**. They provide secure and reliable interconnection to ensure the signal integrity of each data bus. A set of screws are used for fixing the modules, which makes it reliable even in an environment with lots of vibrations.

Using **RAK19009** as your WisBlock Base board, you can make your project compact, which is ideal in small enclosures. You can also use a RAK19005 WisBlock Sensor Extension Cable to position WisBlock Modules apart from the WisBlock Base board or in any part of your case.

Applications

- Wireless Sensor Network
- Environmental monitoring
- Wireless data transmission
- Data acquisition in the industrial environment
- Location and tracking of personnel or moving objects

Features

- Flexible building block design, which enables modular function realization and expansion
- High-speed interconnection connectors in the WisBlock Base board to ensure the signal integrity
- Supports multiple types of low-power MCUs
- Supports multiple types of sensors - a single board can support a combination of two different types of sensors
- Fulfills Industrial level design
- Compact size: 30 × 37 mm

Specifications

Board Overview

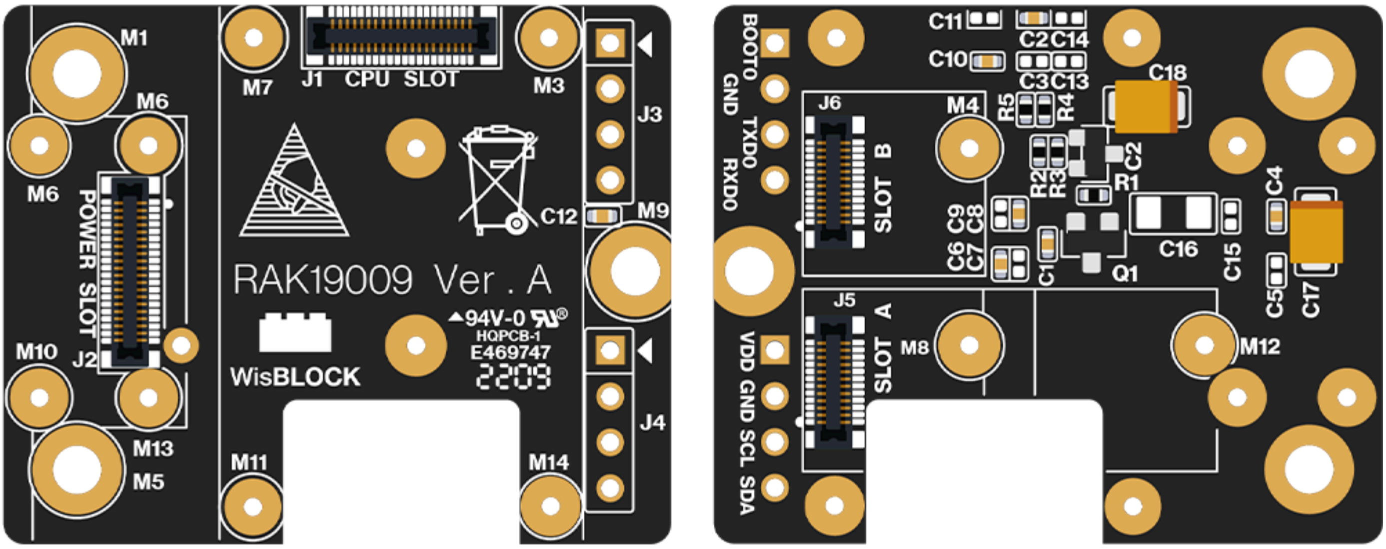

Figure 1: RAK19009 WisBlock Mini Base Top (Left) and Bottom (Right) View

Block Diagram

The block diagram in **Figure 2** shows the internal architecture and external interfaces of the RAK19009 board.

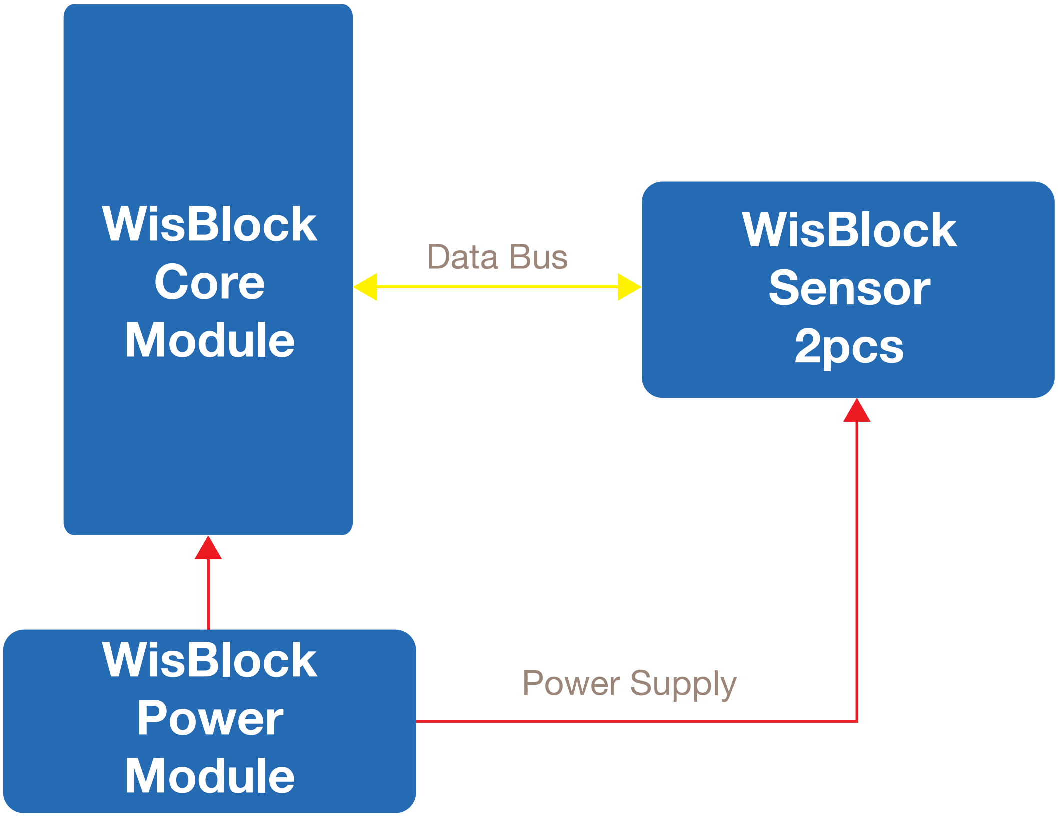

Figure 2: RAK19009 Mini Base block diagram

Hardware

The hardware specification is categorized into six parts. It discusses the interfacing, pinouts, and their corresponding functions and diagrams. It also covers the electrical, mechanical, and environmental parameters that include the tabular data of the functionalities and standard values of the RAK19009 WisBlock Mini Base.

Interfaces

RAK19009 WisBlock Mini Base provides the following interfaces, headers, a button, and WisConnectors:

- 1 WisBlock Core module
- 1 Wisblock Power module
- 2 WisBlock Modules compatible with Slot C-D
- 2 Headers with BOOT, I2C, and UART pins accessible with solder contacts

Figure 3 and **Figure 4** show the location of RAK19009 main components.

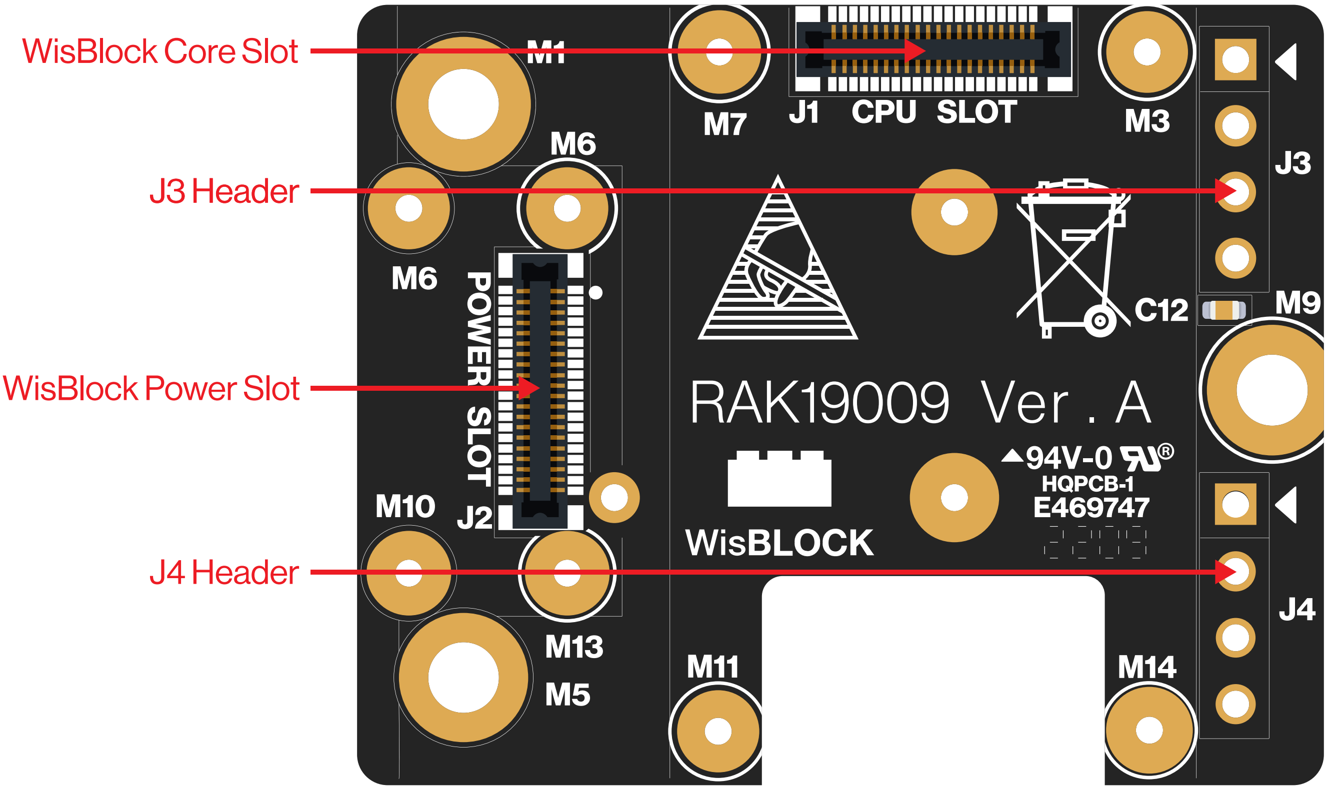

Figure 3: RAK19009 Top View Components

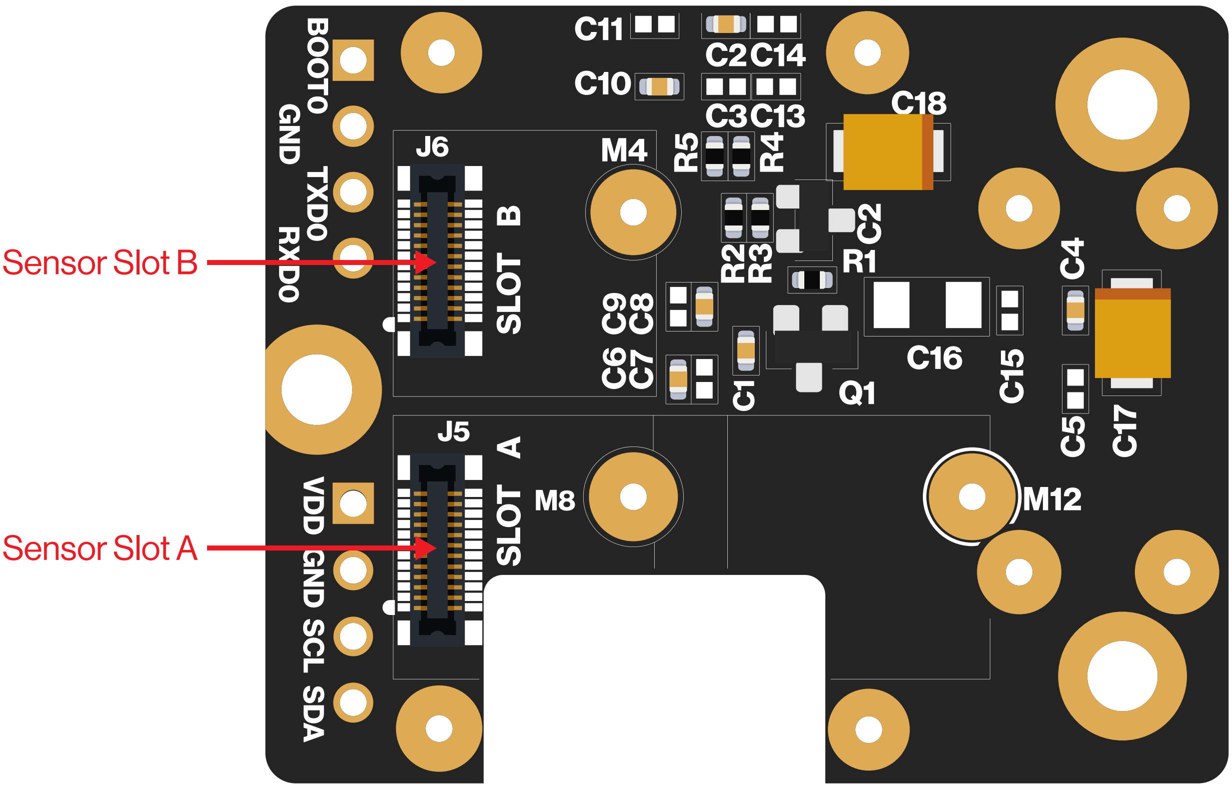

Figure 4: RAK19009 Bottom View Components

J3 and J4 Headers

On the WisBlock Mini Base board, there are two 2.54 mm pitch headers for the IO extension. BOOT, I2C, and UART pins from the WisBlock Core module are also connected to these headers.

J3 Header Pinout

Pin	Pin Name	Description
1	BOOT	MCU Boot pin
2	GND	Ground pin
3	TX0	UART0 Tx pin
4	RX0	UART0 Rx pin

J4 Header Pinout

Pin	Pin Name	Description
1	VDD	3.3 V
2	GND	Ground pin
3	SCL	I2C1 Clock
4	SDA	I2C1 Data

NOTE

BOOT pin is used on startup configuration or sequence of the WisBlock Core connected to it. It is commonly used for uploading the bootloader and/or application firmware. The requirements of the state of the BOOT pin depend on the specific model of the WisBlock Core used.

Pin Definition

Connector for WisBlock Core

The **WisBlock Core module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

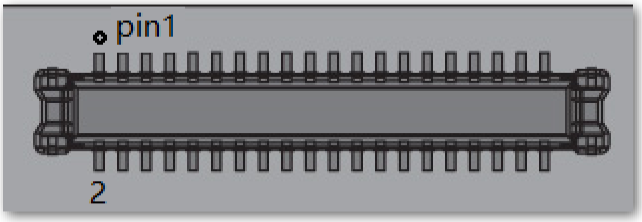

Figure 5: WisBlock Core module connector

The table below shows the pinout of the 40-pin MCU module connector:

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
1	VBAT	2	VBAT
3	GND	4	GND
5	3V3	6	3V3
7	USB+	8	USB–
9	VBUS	10	SW1
11	TXD0	12	RXD0
13	RESET	14	LED1

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
15	LED2	16	IO8
17	VDD	18	VDD
19	I2C1_SDA	20	I2C1_SCL
21	AIN0	22	AIN1
23	BOOT0	24	IO7
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	IO1	30	IO2
31	IO3	32	IO4
33	TXD1	34	RXD1
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND	40	GND

As for the following table, it shows the definition of each pin of the WisBlock Core connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	USB VBUS
10	SW1	I/O	Not connected
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	IO8	I/O	Not connected

Pin Number	Pin Name	Type	Description
17	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
18	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signals
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	BOOT0	I	For ST MCU only. The MCU will enter boot mode if this pin is connected to VDD.
24	IO7	I/O	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock signal
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable

Pin Number	Pin Name	Type	Description
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 RX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connector for WisBlock Power

The **WisBlock Power module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

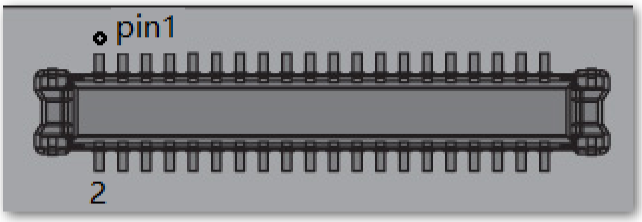

Figure 6: WisBlock Power module connector

The table below shows the pinout of the 40-pin Power module connector:

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
1	VBAT	2	VBAT
3	GND	4	GND
5	3V3	6	3V3
7	USB+	8	USB–
9	VBUS	10	VBUS
11	NC	12	NC
13	RESET	14	LED1
15	LED2	16	NC
17	3V3	18	3V3
19	I2C1_SDA	20	I2C1_SCL
21	AIN0	22	AIN1
23	NC	24	NC
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	NC	30	NC
31	NC	32	NC

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
33	NC	34	NC
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND	40	GND

As for the following table, it shows the definition of each pin of the WisBlock Power connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB–	I/O	USB D–
9	VBUS	S	USB VBUS
10	VBUS	S	USB VBUS

Pin Number	Pin Name	Type	Description
11	NC	NC	Not connected
12	NC	NC	Not connected
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	NC	NC	Not connected
17	3V3	S	3.3 V power supply
18	3V3	S	3.3 V power supply
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signal
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connected
24	NC	NC	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock signal

Pin Number	Pin Name	Type	Description
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	NC	NC	Not connected
30	NC	NC	Not connected
31	NC	NC	Not connected
32	NC	NC	Not connected
33	NC	NC	Not connected
34	NC	NC	Not connected
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connectors for WisBlock Sensor

The WisBlock sensor module connector is a **24-pin board-to-board connector**.

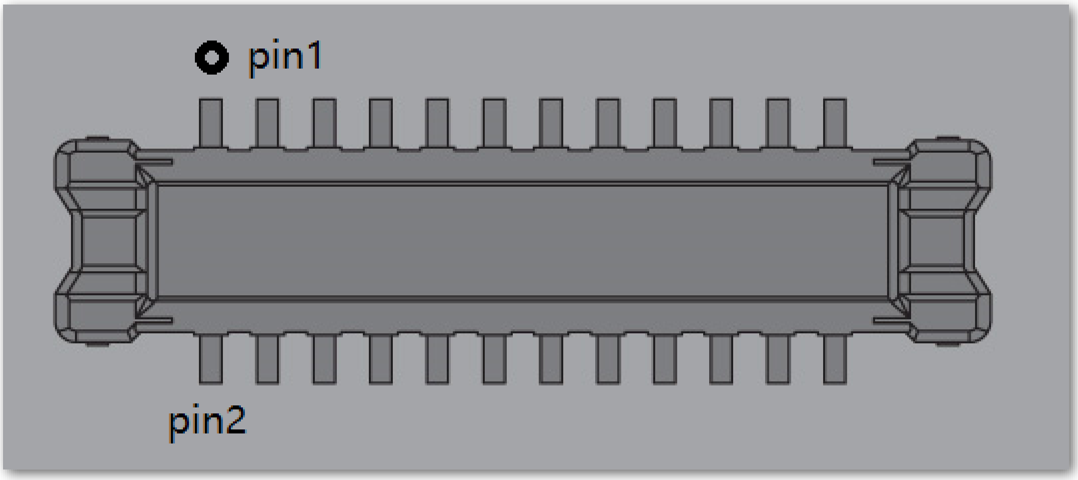

Figure 7: WisBlock Sensor module connector

NOTE

There are two connectors reserved for the sensor modules on the RAK19009. The pinout definition of the WisBlock modules with a 24-pin connector on WisBlock Mini Base varies according to its connector.

Connector C	Connector D	Pin Number	Pin Number	Connector C	Connector D
TXD1	TXD1	1	2	GND	GND
SPI_CS	SPI_CS	3	4	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	5	6	SPI_MOSI	SPI_MOSI
I2C1_SCL	I2C1_SCL	7	8	I2C1_SDA	I2C1_SDA
VDD	VDD	9	10	IO4	IO6
3V3_S	3V3_S	11	12	IO3	IO5
NC	NC	13	14	3V3_S	3V3_S
NC	NC	15	16	VDD	VDD

Connector C	Connector D	Pin Number	Pin Number	Connector C	Connector D
NC	NC	17	18	NC	NC
NC	NC	19	20	NC	NC
NC	NC	21	22	NC	NC
GND	GND	23	24	RXD1	RXD1

As for the following table, it shows the pin name and description of each pin in the WisBlock Sensor module connector.

Pin Number	Connector C	Connector D	Type	Description
1	TXD1	TXD1	I/O	UART TX signal
2	GND	GND	S	Ground
3	SPI_CS	SPI_CS	I/O	SPI chip select signal
4	SPI_CLK	SPI_CLK	I/O	SPI clock signal
5	SPI_MISO	SPI_MISO	I/O	SPI MISO signal
6	SPI_MOSI	SPI_MOSI	I/O	SPI MOSI signal
7	I2C1_SCL	I2C1_SCL	I/O	I2C clock signal
8	I2C1_SDA	I2C1_SDA	I/O	I2C data signal
9	VDD	VDD	S	Generated by CPU module. Used to power sensor board if MCU IO level is not 3.3 V

Pin Number	Connector C	Connector D	Type	Description
10	IO4	IO6	I/O	General purpose IO pin. When 3V3_S is used, this pin cannot be used as an interrupt input.
11	3V3_S	3V3_S	S	3.3 V power supply. This power pin is controlled by IO2 from the WisBlock Core module.
12	IO3	IO5	I/O	General purpose IO pin. When 3V3_S is used, this pin cannot be used as an interrupt input.
13	NC	NC	NC	Not connected
14	3V3_S	3V3_S	S	3.3 V power supply. This power pin is controlled by IO2 from the WisBlock Core module.
15	NC	NC	NC	Not connected
16	VDD	VDD	S	Generated by CPU module. Used to power sensor board if the MCU IO level is not 3.3 V.
17	NC	NC	NC	Not connected
18	NC	NC	NC	Not connected
19	NC	NC	NC	Not connected
20	NC	NC	NC	Not connected

Pin Number	Connector C	Connector D	Type	Description
21	NC	NC	NC	Not connected
22	NC	NC	NC	Not connected
23	GND	GND	S	Ground
24	RXD1	RXD1	I/O	UART RX signal

Electrical Characteristics

Absolute Maximum Ratings

The **Absolute Maximum Ratings** of the device are shown in the table below. The stress ratings are the functional operation of the device.

WARNING

1. If the stress rating goes above what is listed, it may cause permanent damage to the device.
2. Under the listed conditions is not advised.
3. Exposure to maximum rating conditions may affect the device reliability.

Ratings	Maximum Value	Unit
IOs of WisConnector	−0.3 to VDD+0.3	V
ESD	2000	V

NOTE

The RAK19009, like any electronic equipment, is sensitive to **electrostatic discharge (ESD)**. Improper handling can cause permanent damage to the module.

Current Consumption

The RAK19009 is designed for **low-power IoT products**. It has no power interface, so it must be used with a WisBlock Power module. When there is no module on RAK19009, the **leakage current is lower than 2 µA**. With WisBlock Core and WisBlock sensor on it, the sleep current is **lower than 10 µA**. When a LoRa module is transmitting, the current may reach **130 mA**.

Conditions	Current	Unit
Leakage current, without any module on RAK19009	2	µA
Idle current, with WisBlock Core and WisBlock Modules in sleep mode	10	µA
Working current, with LoRa module transmitting	130	µA
Maximum output current	750	mA

Mechanical Characteristics

Board Dimensions

NOTE

You may also refer and download the [M1.2 Stand-off fastener/inserts datasheet](#) .

Figure 8 shows the detailed mechanical dimensions of RAK19009.

Room 506, Bldg B, New Compark, Pingshan First Road, Taoyuan Street, Nanshan District, Shenzhen, Guandong Province, The People's Republic of China

www.RAKwireless.com

+86-755-86108311

inquiry@RAKwireless.com

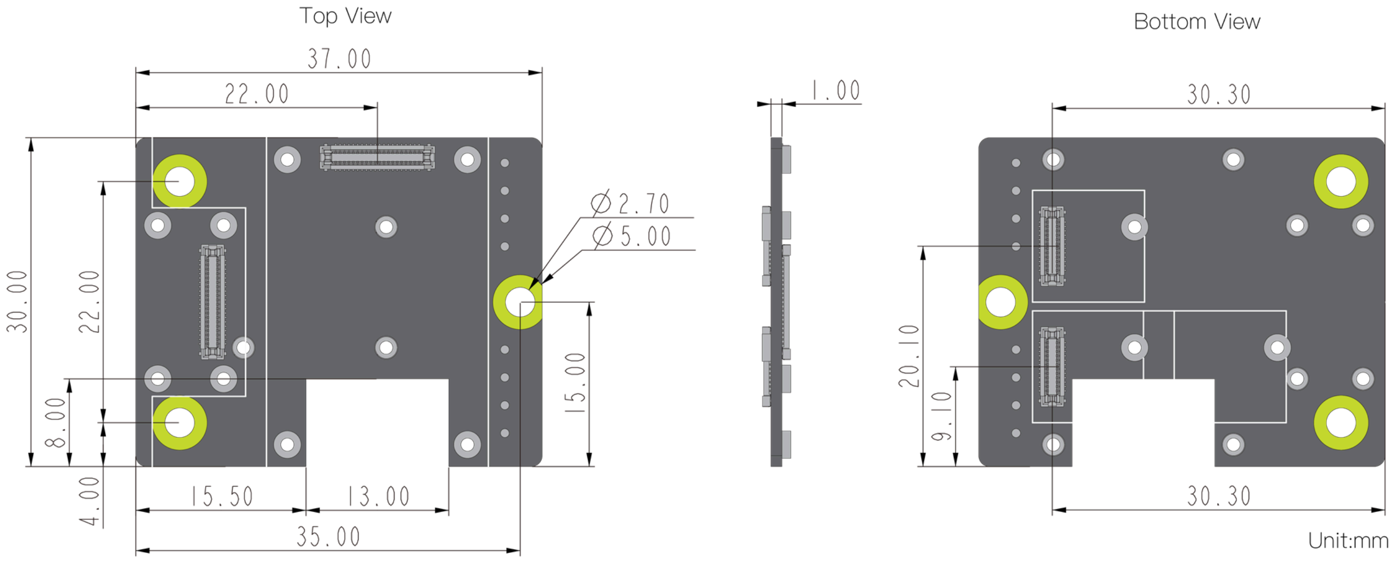

Figure 8: RAK19009 mechanical dimensions

Figure 9 show the mounting holes location and diameter of the RAK19009 Board.

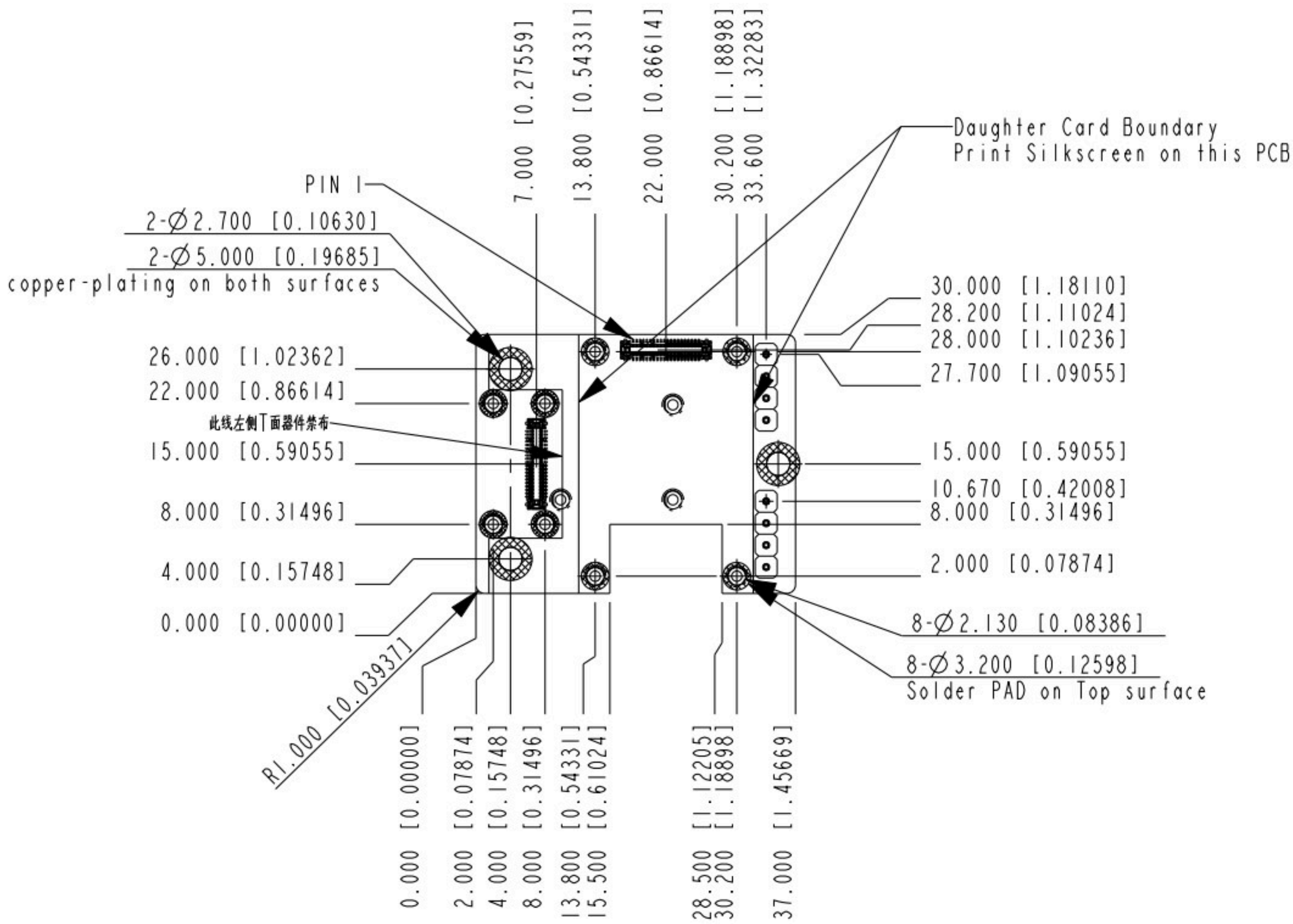

Figure 9: RAK19009 mounting holes location and diameter

WisConnector PCB Layout

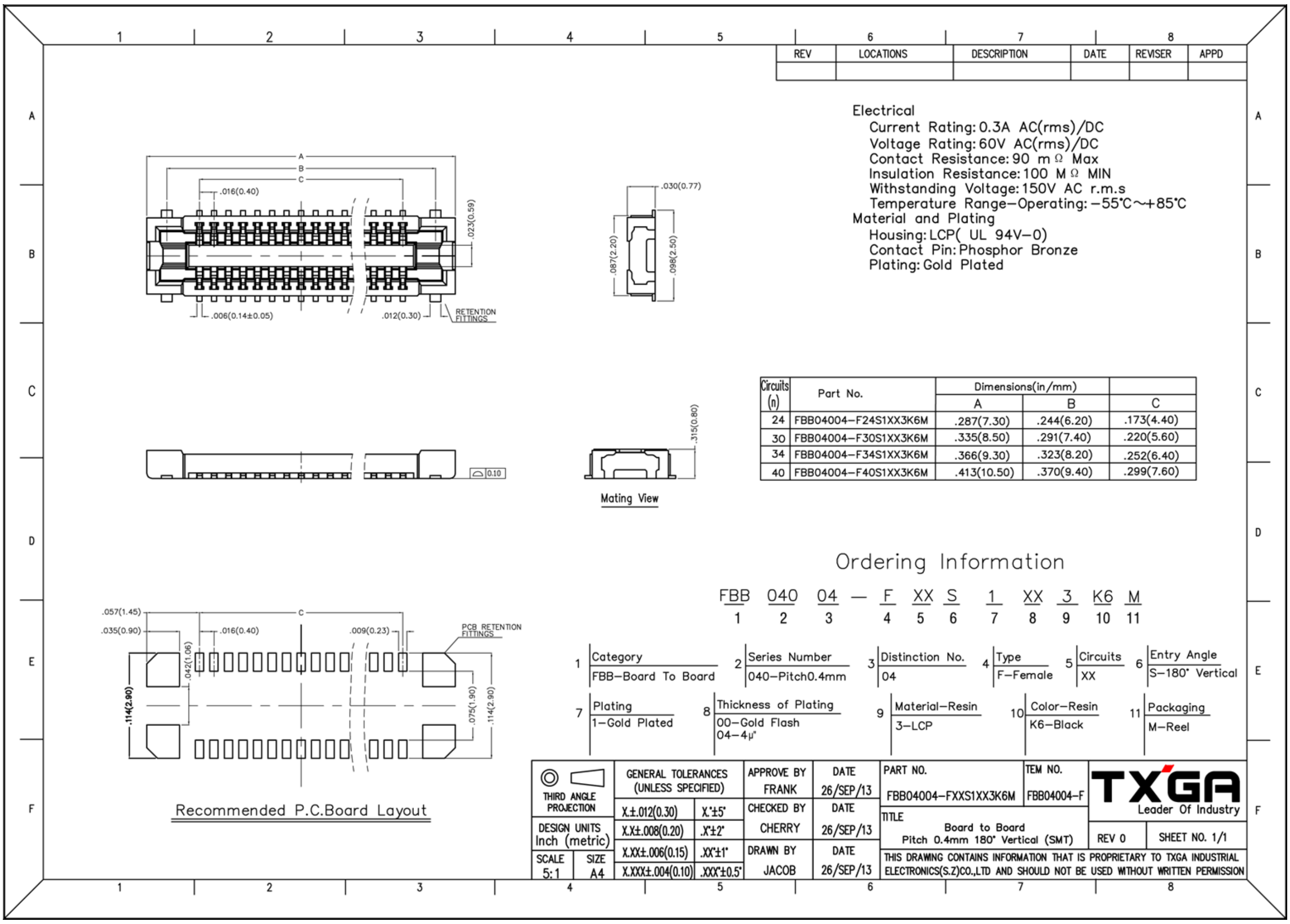

Figure 10: WisConnector PCB footprint and recommendations

Environmental Characteristics

The table below lists the operation and storage temperature requirements of RAK5005-O:

Parameter	Minimum	Typical	Maximum
Operational temperature range	−35 °C	+25 °C	+75 °C
Extended temperature range	−40 °C	+25 °C	+80 °C
Storage temperature range	−40 °C	+25 °C	+80 °C

Schematic Diagram

The component schematics diagram of the RAK19009 is shown in **Figure 11**.

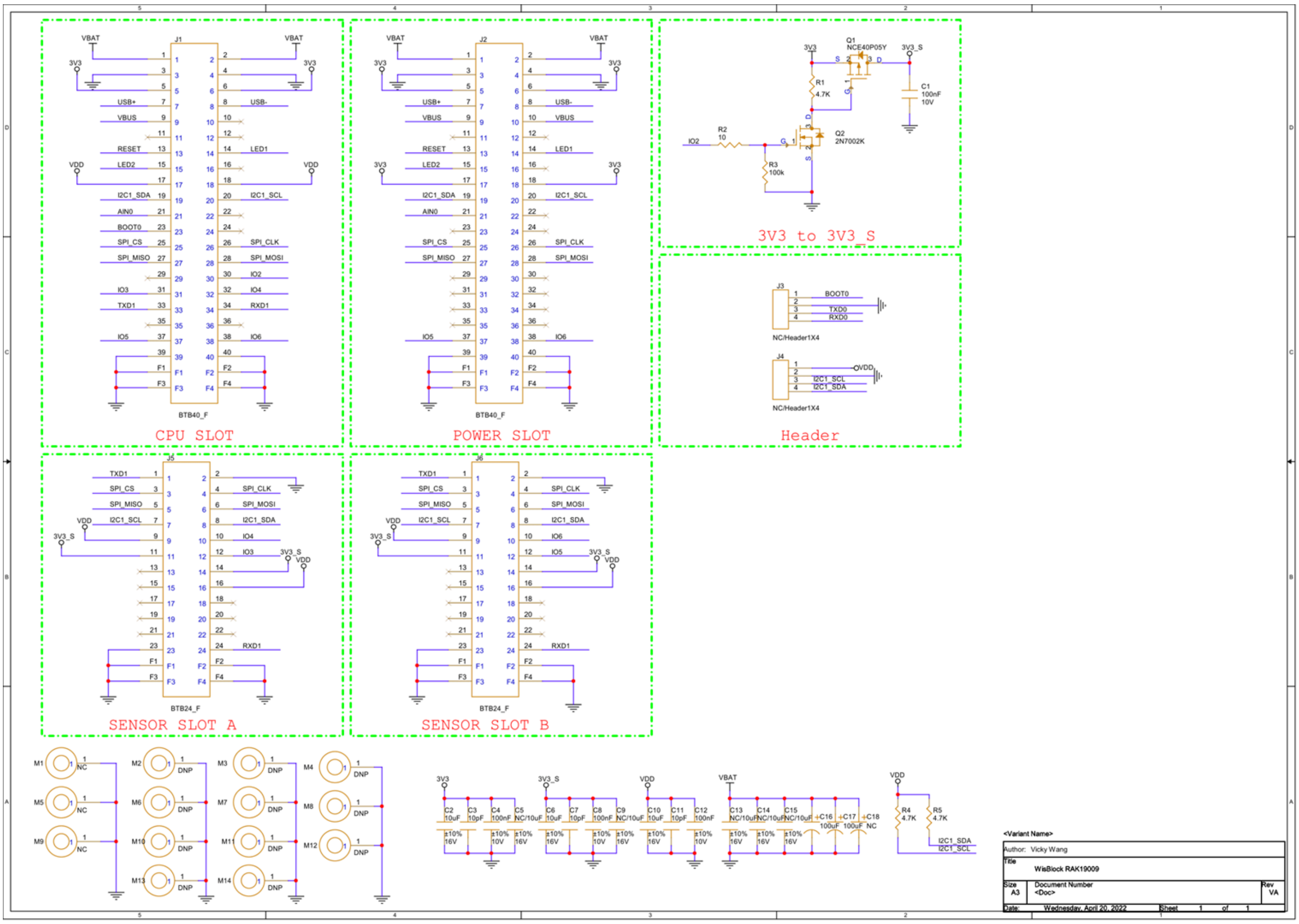

Figure 11: RAK19009 Schematic Diagram